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Chapter 7. Exit nodes, the Mullvad add-on, and what I run

> **The payoff of this chapter:** the routing rule that governs VPNs on the road, the setup I use (a standalone Mullvad account that I can switch on and off, giving up homelab access while it's on), what a Tailscale **exit node** is, and the paid add-on that *can* give you privacy and your homelab at the same time, if you're willing to trade control and a few more dollars for it.

At home, the pieces share the Pi's network. Away from home, **Tailscale** is the link back to the Pi. This part picks up where Chapter 6 left off: the VPN choice is made; now it has to coexist with the homelab.

Reaching your homelab from anywhere (the easy part)

This part already works. Because Tailscale is a mesh VPN, any device signed into the tailnet reaches the Pi from anywhere, coffee shop, cellular, hotel, with no extra setup:

- **Audiobookshelf:** `https://abs.home` streams just as it does at home.
- **The dashboard:** `https://home.home` (or `https://homelab`) opens from anywhere.

That is inbound access over an authenticated tunnel. The hard parts below are outbound: where traffic exits and which DNS resolver answers.

Pi-hole already follows you (set up in Chapter 4)

No extra setup is needed here. When Pi-hole became the tailnet DNS in Chapter 4 (the **Override local DNS** setup), Pi-hole ad-blocking came with it on every tailnet device. On cellular, the phone still resolves through Pi-hole, so ads stay blocked. The same step is also what makes your `.home` names work away from home.

The conflict starts when another VPN is turned on, which is the rest of this chapter.

> **!** This only holds while Tailscale is the active VPN
>
> The whole mechanism rides on Tailscale being the thing controlling your device's DNS. The instant another VPN takes over (next section), this stops applying. That's not a bug, it's the central tension of going mobile, and the rest of this chapter is about navigating it.

The one rule that governs everything off-network

The road setup follows this rule:

> **A full-device VPN owns the default route *and* the DNS, and a device runs only one such VPN at a time.**

Two consequences fall out of it:

1. **A privacy VPN and Tailscale can't both be the active tunnel.** Phones make this a hard wall: iOS and Android allow exactly one active VPN, so turning on Mullvad (or Aura) drops Tailscale. I ran into the same problem on a laptop, for a different reason: **routing.** A Mullvad WireGuard tunnel grabs the entire default route, which swallows the path back to your tailnet, so the moment Mullvad is up, `https://homelab` and your `100.x` addresses go dark there too. The phone enforces "one VPN" by policy; the laptop enforces it by routing.
2. **Whatever VPN is active forces its own DNS.** Even setting the one-VPN limit aside, the active VPN routes DNS through its own resolvers. So while you're on *any* location/privacy VPN, your traffic is *not* going through Pi-hole, by design, to prevent DNS leaks.

> **i** Why two VPN apps don't give you everything at once
>
> This is why "use my own VPN *and* filter through Pi-hole *and* reach my homelab, all at once, on one device" has no clean answer with two separate apps: Pi-hole and your homelab live only on your tailnet, reaching them needs Tailscale to be the active tunnel, and a privacy VPN that's up takes that role (by policy on a phone, by routing on a laptop). The only real escapes are to put the privacy VPN and your homelab on the *same tunnel*, which is the exit-node idea further down, or to switch between them, which is what I do.

What I run: keep them separate, switch per activity

After trying to have it all at once and losing (see the sidebar), I settled on the setup that has held up for me: **I keep a standalone Mullvad subscription and Tailscale installed side by side, and I only ever run one at a time.** On both my laptop and my phone.

In practice that means I segregate my activities into two modes and flip between them in a couple of seconds:

- **Tailscale on, Mullvad off (my default).** Homelab reachable, Pi-hole ad-blocking everywhere via the DNS push, `.home` names resolve. This is where I live almost all the time. The cost is that my real IP is showing.
- **Mullvad on, Tailscale off (when privacy matters).** Sketchy Wi-Fi, or I want to change region or hide my IP. While it's on I accept that the homelab and Pi-hole are unavailable for that stretch, Mullvad carries my DNS and its own ad-blocking instead. When I'm done, Mullvad off, Tailscale back on.

My setup is intentionally plain. No exit node, no policy routing, no add-on. The downside is right there in the open: **I can't reach my homelab while the privacy VPN is on**, so I plan around it. For me that beats the alternatives, because I keep full control of my own Mullvad subscription, its config files, its settings, and I'm not paying for anything extra.

> **i** I tried hard to have both on one box, and couldn't make it stick
>
> Before settling on switch-per-activity, I spent hours and hours trying to run Mullvad and Tailscale *simultaneously* on my laptop. The approaches, and why each fell over:
>
> - **Split tunneling / carving out the tailnet.** Mullvad's WireGuard config claims the entire default route (`AllowedIPs = 0.0.0.0/0`), which also swallows the return path to Tailscale's range (`100.64.0.0/10`). The plan was to exclude that range from the Mullvad tunnel so my devices could still reach the homelab.
> - **Forcing Tailscale's route priority** so its `100.x` routes would win over Mullvad's catch-all default.
>
> I could occasionally get a fragile version working, but never one that survived a reconnect, a Mullvad server change, or a reboot. It was never reliable enough to trust. The lesson I took: with two separate full-device VPN apps, having both on at once is not a stable setup. If you want both at the same time, change the architecture; that's the exit-node add-on below.

(Unverified by me) The thing that *would* let you have both: an exit node

Everything above assumes two separate VPN apps. There's a different architecture that apparently sidesteps the whole "one tunnel at a time" problem, and it's good to know even if, like me, you don't use it yet.

Normal Tailscale is an **overlay network**: it only carries traffic *between* your own devices (laptop `homelab` phone). Your ordinary internet traffic still leaves through your local connection; Tailscale doesn't touch it.

An **exit node** changes that. You designate one device on your tailnet as "the exit," and your other devices route **all** their internet traffic through it, the full default route, not just tailnet traffic. To the outside world, your traffic now appears to come from the exit node's IP. That's exactly how a traditional full-tunnel VPN behaves, except the "VPN server" is a node on your *own* tailnet, so **you're still on Tailscale the whole time**. Only one exit node is active per device, and you switch it off by selecting "None":

```
tailscale exit-node list                  # list available exit nodes
tailscale set --exit-node=<node>          # route everything through it
tailscale set --exit-node=                # turn it back off
```

Because an exit node *is* still Tailscale, choosing one doesn't drop your tailnet. Your homelab stays reachable while your traffic exits somewhere else. That architecture escapes the one-rule trap, and it comes in three flavors.

Flavor 1 (the upgrade path): the Tailscale Mullvad add-on

This is the option that buys you privacy *and* your homelab at the same time, and it's the thing I'd reach for if my needs grow. Tailscale sells access to **Mullvad's worldwide WireGuard servers as a paid add-on purchased through Tailscale**. Mullvad's servers appear as selectable exit nodes inside the tailnet:

```
tailscale exit-node list                   # Mullvad cities now appear here
tailscale set --exit-node=<mullvad-node>   # exit through Mullvad, stay on
Tailscale
```

On **iOS / macOS / Android**: the **•••** menu → **Use exit node** → pick a Mullvad **location**. Because it's still Tailscale, your homelab and `.home` names keep working while your traffic exits through Mullvad. That gives you both pieces: always-on privacy that doesn't cost you the homelab, and it's the only option that works cleanly on a phone without fighting the one-VPN rule.

So why don't I run it? Because of what you give up:

- **Cost:** about **$5/month** on top of things. If you also keep your own Mullvad subscription (as I do, for the control), you're now at **$10+/month** for the pair. The add-on alone (~$5) can *replace* your own sub, but then you lose the next point entirely.
- **Control.** When you buy the add-on, **Tailscale provisions a Mullvad account for you that you have very little say over.** You don't get your own Mullvad login, you can't download WireGuard config files to run on the Pi or a VPS, and you can't tune Mullvad's own settings. You get exit nodes and nothing else. For someone who specifically wants to *own and control* their Mullvad config (which is the whole reason Chapter 6 pointed at Mullvad), that's a real loss.

Convenience versus control

It comes down to convenience versus control. The add-on gives you *convenience* (one tunnel, always on, homelab intact). Owning your own Mullvad sub gives you *control* (portable configs, full settings, no extra account you don't manage). Right now I value control more, and I'm fine switching apps by hand to get it. If my needs shift toward "always-on privacy that never costs me the homelab," I'll pay the extra and run both, or move fully to the add-on. It's a budget-and-priorities call, not a right-or-wrong one.

When you *do* use a Mullvad exit node, two flags matter on the road:

```
# Keep access to LAN devices (printers, etc.) while using an exit node
tailscale set --exit-node=<mullvad-node> --exit-node-allow-lan-access
```

- **`--exit-node-allow-lan-access`**: by default, turning on *any* exit node cuts your access to the local network you're physically on. This flag restores it. On mobile it's the **Allow LAN access** toggle when you pick the exit node.
- **DNS leaks:** allowing LAN access can let some DNS queries escape the tunnel. Recent Tailscale clients (1.48.3+) route DNS correctly through Mullvad exit nodes without extra config, so keep your clients updated.

Flavor 2: your own Pi as an exit node (free, but not private)

You can advertise one of your *own* machines as an exit node. The helper script in this chapter's folder does it on the Pi:

```
sudo tailscale set --advertise-exit-node
# then approve it: admin console -> Machines -> the Pi ->
#   Edit route settings -> enable "Use as exit node"
```

On Linux you also enable IP forwarding (`net.ipv4.ip_forward=1` and the IPv6 equivalent); the script sets both.

- **What it's good for:** routing through your **home** Pi so you appear to be at home, reaching home-only services, or content tied to your home connection, and, importantly, it's the one exit option that keeps your traffic on your own network where **Pi-hole still applies**. This is the sweet spot when what you want on the road is ad-blocking, not IP-hiding.
- **What it's not:** privacy. Traffic exits from **your own home IP**, so there's no hiding. And don't use a *phone* as an exit node, it routes in userspace and is slow; the Pi (kernel routing) is fine.

Flavor 3 (advanced): a Pi-style exit node chained through Mullvad

Flavor 2 hides nothing, because the box exits from the home IP. You can fix that by *chaining*: run an exit node that itself tunnels out through a standalone Mullvad **WireGuard** config. Your devices reach the node over the tailnet, and the node forwards their traffic out through Mullvad:

```
you -> Tailscale -> your exit-node box -> Mullvad WireGuard -> internet
```

This is the only way to use a **standalone** Mullvad subscription *as a Tailscale exit node* without paying for the add-on, so it keeps the control you wanted while still giving you the always-on-plus-homelab shape. The shape of it:

1. Use a machine that is **not your home Pi** if privacy is the goal (a cheap Linux VPS works well). The home Pi only makes sense for "appear at home," where adding Mullvad would defeat the purpose.
2. On that box, install Tailscale, advertise it as an exit node, and enable IP forwarding (same as Flavor 2).
3. Download a **WireGuard config** from your Mullvad account and bring it up (`wg-quick up <conf>`) so the box's default route leaves through Mullvad.
4. The sharp edge: Mullvad's config claims the **entire** default route (`AllowedIPs = 0.0.0.0/0`), which also swallows the return path to your tailnet. You have to keep Tailscale's range (`100.64.0.0/10`) out of the Mullvad tunnel so your devices can still reach the node. That policy-routing fiddliness is exactly the same fight I lost on the laptop, just moved to a box you can leave running, and it's the work the add-on saves you.

I don't know if this would actually work, since I basically spent hours trying to make something very similar work on my RPi. I know that it's technically different since that RPi is my DNS server on my router and for tailscale and I guess that makes it all more complicated, but I'm skeptical this would work.

A router-level VPN might be another answer, but I can't test it yet

A friend with 10+ years of homelab experience recently pointed out another possible path: configure the **router itself** to connect to a VPN. In theory, that could move the privacy tunnel out of the phone/laptop app layer and into the network layer, which may avoid some of the conflicts in this chapter.

I can't validate that setup on my current hardware. My router and its firmware are too old and do not expose the VPN-client features needed to try it. He also doesn't use Mullvad, so I don't know yet whether this works cleanly with Mullvad's WireGuard configs or whether it introduces a different set of routing or DNS problems.

So treat all of these things as a lead, not a recommendation: if your router supports acting as a WireGuard/OpenVPN client, it may be worth investigating. It is not part of the tested setup in this guide.

The decision matrix

You can't have all three of {homelab access, Pi-hole filtering, privacy VPN} on one device at one time with two separate apps. But you *can* switch between coherent modes in seconds. Pick per situation:

You're away and you want…	What you turn on	Pi-hole?	Homelab?	Hides IP?
Homelab + ad-blocking (my default)	Tailscale on, no exit node	Yes	Yes	No
Privacy, my way (what I run)	Mullvad on, Tailscale off	No	No	Yes
Appear at home / reach LAN	Tailscale + Pi exit node (Flavor 2)	Yes	Yes	No
Privacy *and* homelab at once	Mullvad add-on exit node (Flavor 1)	No	Yes	Yes

The pattern to read out of that table: with two separate apps (rows 1 and 2) you trade the whole homelab for privacy and switch between them, which is cheap, fully under your control, and what I do. The exit-node rows (3 and 4) are the only ways to keep the homelab *while* your traffic exits elsewhere, and the only one of those that also hides your IP is the paid Mullvad add-on. Two things the Yes/No columns gloss over: in the Mullvad rows your DNS goes to Mullvad, so Pi-hole is bypassed and you lean on Mullvad's own ad-blocking instead; and "appear at home" hides nothing because traffic exits from the home IP.

A practical everyday routine

For most people the comfortable default is simple, and it's mine:

1. **Leave Tailscale on, no exit node, all the time.** Homelab reachable, Pi-hole ad-blocking everywhere via the DNS push, phone behaves normally. You give up IP-hiding, which most of the time you don't need.
2. **When you specifically want privacy**, switch to standalone Mullvad for that session and accept that the homelab steps aside until you switch back. Cheap, reliable, fully yours.
3. **If "always-on privacy without ever losing the homelab" becomes worth the money and the loss of control**, buy the Mullvad add-on (Flavor 1) and let a Mullvad exit node do both at once, or stand up a chained exit node (Flavor 3) if you'd rather keep your own config.

Recap: the whole series

- **Reaching your homelab away** is the free win: Tailscale makes `homelab:13378` and `https://home.home` work from anywhere, no extra setup.
- **Pi-hole on the road** rides on the Chapter 4 DNS push, and works only while Tailscale is your active VPN.

- **The governing rule:** a full-device VPN owns the route and DNS, and a device runs one at a time (by policy on a phone, by routing on a laptop), so a separate privacy VPN and your homelab can't both be live with two apps.
- **What I run:** a standalone Mullvad I keep fully under my own control, switched on only when I want privacy, accepting that the homelab is offline while it's on. Boring, reliable, cheap.
- **The escape hatch, when you want both at once:** an **exit node**, most easily the paid **Mullvad add-on**, which trades some control and a few more dollars for always-on privacy that never costs you the homelab.

With that, your homelab is fully mobile: an always-on Raspberry Pi that streams your media, blocks ads on every device, presents a single `https://home.home` control panel, and gives you a deliberate, switchable answer for privacy on the road. All of it private by default, over one Tailscale network, with nothing exposed to the public internet.

That completes the core of the homelab. **Volume III** is the extras: getting a *phone and Linux machines* to talk to each other for everyday remoting, file transfer, and clipboard sharing, starting with Chapter 8.